A Multifaceted Analysis of Sri lankan Payment Adoption, Borrowing Behavior, and Financial Clustering

Progress Presentation

Group -04

AGENDAS

- Project Overview
- Introduction of the problem
- Objectives
- Literature Review
- Progress
- Future Works
- Conclusion
- References

OUR TEAM

- PS/2021/078 M.C.Madhusanka
- PS/2021/027 K.A.G.D.Marapana
- PS/2021/245 S.P.Wanigasingha
- PS/2021/207 H.M.C.J.Bandara
- PS/2021/219 H.M.K.J.Herath
- PS/2021/173 J.A.A.R.Jayaweera
- PS/2021/164 R.R.A.Indranath
- PS/2021/005 J.A.D.D.S. JAYASOORIYA
- PS/2021/114 W.J.E.S.V Weerasinghe
- PS/2021/103 K.G.R.Kaveesha

PROJECT OVERVIEW

"Digital Payment Adoption Analysis"

"Borrowing Behavior Patterns"

"Financial Inclusion Mapping"

"Population Segmentation via Clustering"

📍 **SCOPE: Sri Lanka's Financial Ecosystem**

📊 **DATA: Global Findex 2025 (1000 Sri lankan respondents)**

🎯 **FOCUS: Digital payment adoption, borrowing patterns, financial inclusion**

🔄 **APPROACH: Mixed-methods data analysis and machine Learning**

📈 **OUTPUT: Actionable insights for policymakers & financial institutions**

Introduction of the Problem

- Many Sri Lankans still struggle to access modern financial services.
- There are gaps in digital payments, borrowing opportunities, and financial inclusion, especially across age, income, and rural groups.
- This research explores these issues and identifies patterns to support better solutions.

Key Problems

📌 Low Digital Adoption — Limited access, trust issues, and skills gaps

📌 Unequal Credit Access — Rural & low-income groups depend on informal loans

📌 Financial Exclusion — Many lack bank accounts or saving opportunities

📌 No Targeted Segmentation — Services aren't tailored to real customer needs

68% Progress

Research Objectives

Four Pillars of Our Investigation

🎯 *OBJECTIVE 1: MAP DIGITAL PAYMENT ADOPTION*

- Quantify adoption rates across demographics
- Identify key barriers and enablers
- Measure the digital divide

🎯 *OBJECTIVE 2: ANALYZE BORROWING BEHAVIORS*

- Catalog borrowing sources (formal vs informal)
- Understand borrowing purposes
- Assess credit access inequalities

🎯 *OBJECTIVE 3: PROFILE FINANCIAL INCLUSION*

- Measure account ownership disparities
- Analyze saving behaviors
- Evaluate financial resilience

🎯 *OBJECTIVE 4: SEGMENT THE POPULATION*

- Identify distinct financial behavior clusters
- Develop targeted intervention strategies
- Create persona-based policy recommendations

Literature Review

We build our study on these established findings to confirm, challenge, and explore the "why" in Sri Lanka's context.

1. The "Access vs. Usage" Gap

Known: High bank account access does not guarantee active digital payment use.
Our Goal: Quantify this gap and identify if the main barriers are trust, cost, usability, or simply no perceived need.

2.The "Mid-Life" Borrowing Peak

Known: Borrowing often peaks in the 35-44 age group due to life-cycle needs.
Our Goal: Verify this "inverted U" pattern in our data and link loan purposes (e.g., business vs. health) to different age groups.

3.Behavior Trumps Infrastructure

Known: Urban-rural digital gaps can be small, implying behavior and habit are bigger barriers than infrastructure.
Our Goal: Compare adoption rates between groups and study if small, frequent digital transactions can build lasting habits.

Literature Review Summary & Research Gap

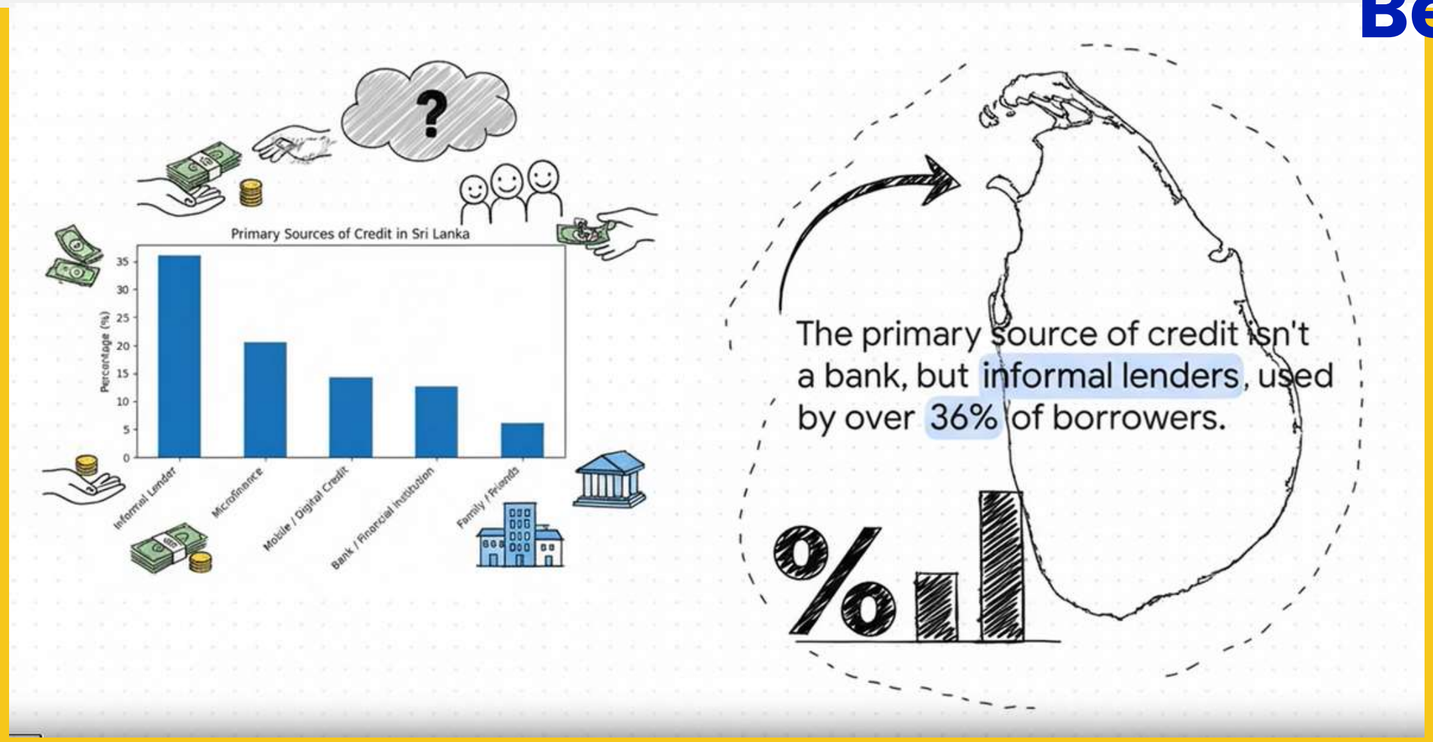
Topic	**What Literature Says** (Findings)	**Identified Gap** in Research	**Your Research Focus** (Contribution)
Digital Payments	Usage increasing; barriers include risk, lack of trust, UX	Limited Sri Lanka-specific demographic studies	Predict adoption using demographic & behavioral factors
Borrowing Behavior	Borrowing varies by age, income, and job type	Lack of predictive models in Sri Lanka context	Build models to predict borrower vs non-borrower
Financial Inclusion	Mobile money improves access; usage still uneven	Few SL studies on usage & savings behavior	Analyze accessibility & saving trends in Sri Lankan context
Cluster Analysis	Clustering reveals useful financial behavior segments globally	Almost no clustering work focused on Sri Lanka	Create Sri Lankan financial behavior segments (clusters)

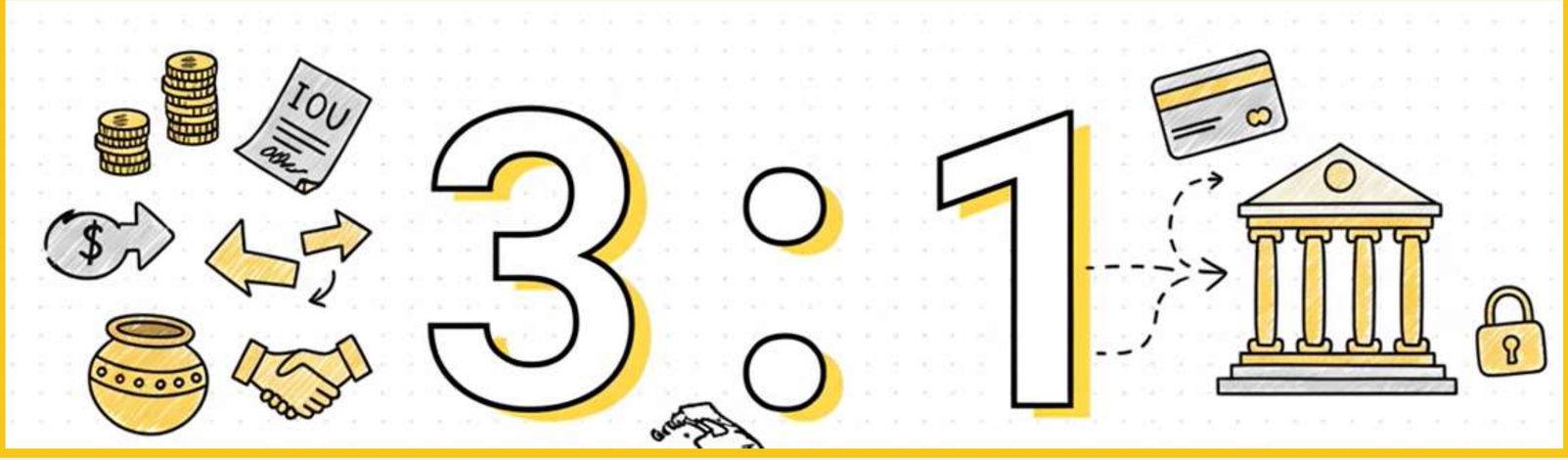
Progress - Overall Status

Component	Status	Completion
Data Acquisition & Cleaning	✅ Complete	100%
Part 1: Digital Payment Adoption	✅ Complete	100%
Part 2: Borrowing Analysis	✅ Complete	100%
Part 3: Financial Inclusion Patterns	🔄 In Progress	60%
Part 4: Clustering Analysis	🔄 In Progress	10%
Integration & Synthesis	⬜ Not Started	0%
Documentation & Reporting	⬜ Not Started	0%

Progress - Key Findings

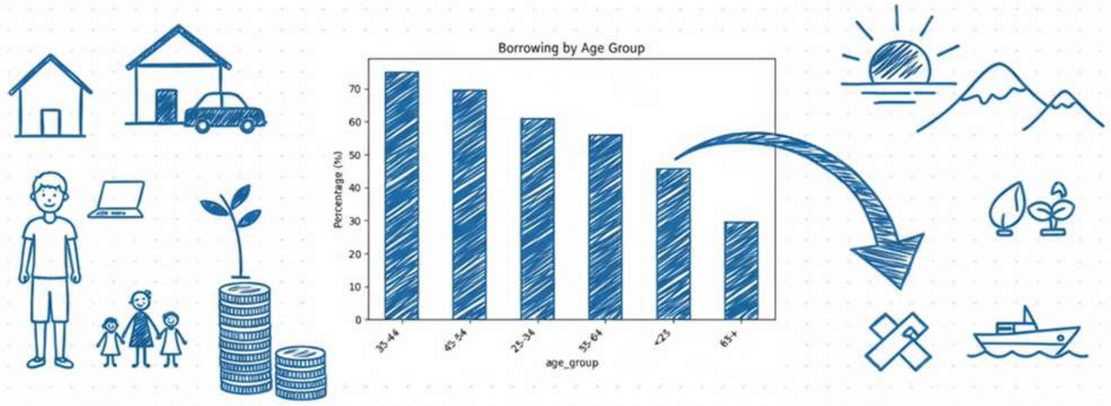
Borrowing Behavior



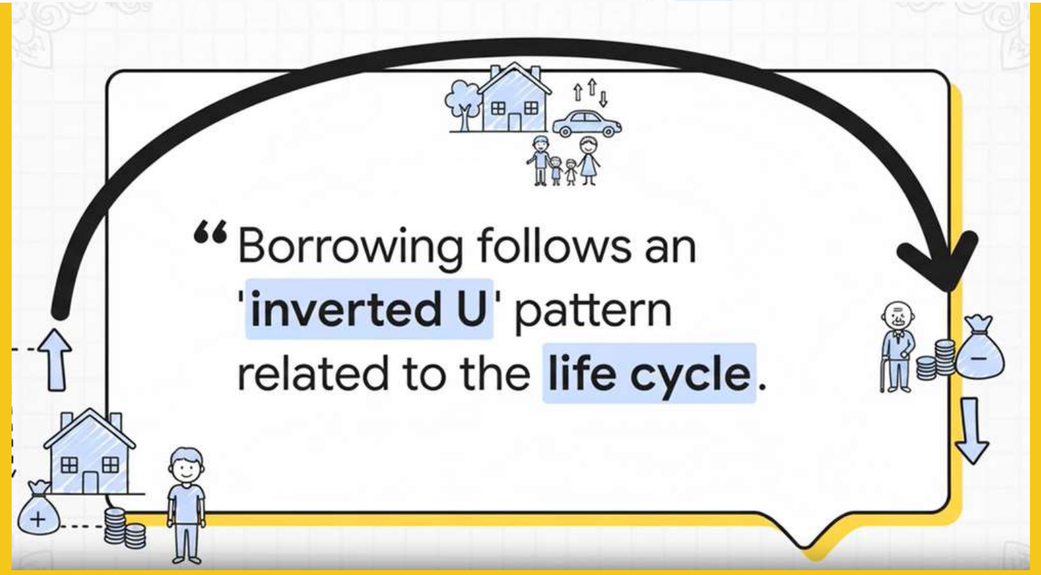


Progress - Key Findings

Borrowing Behavior

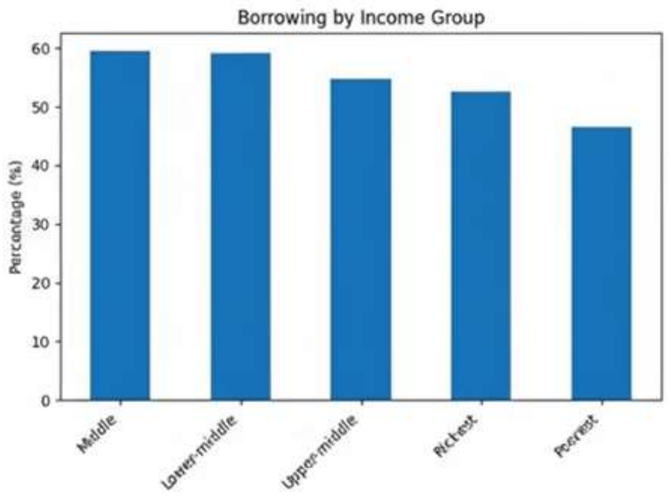


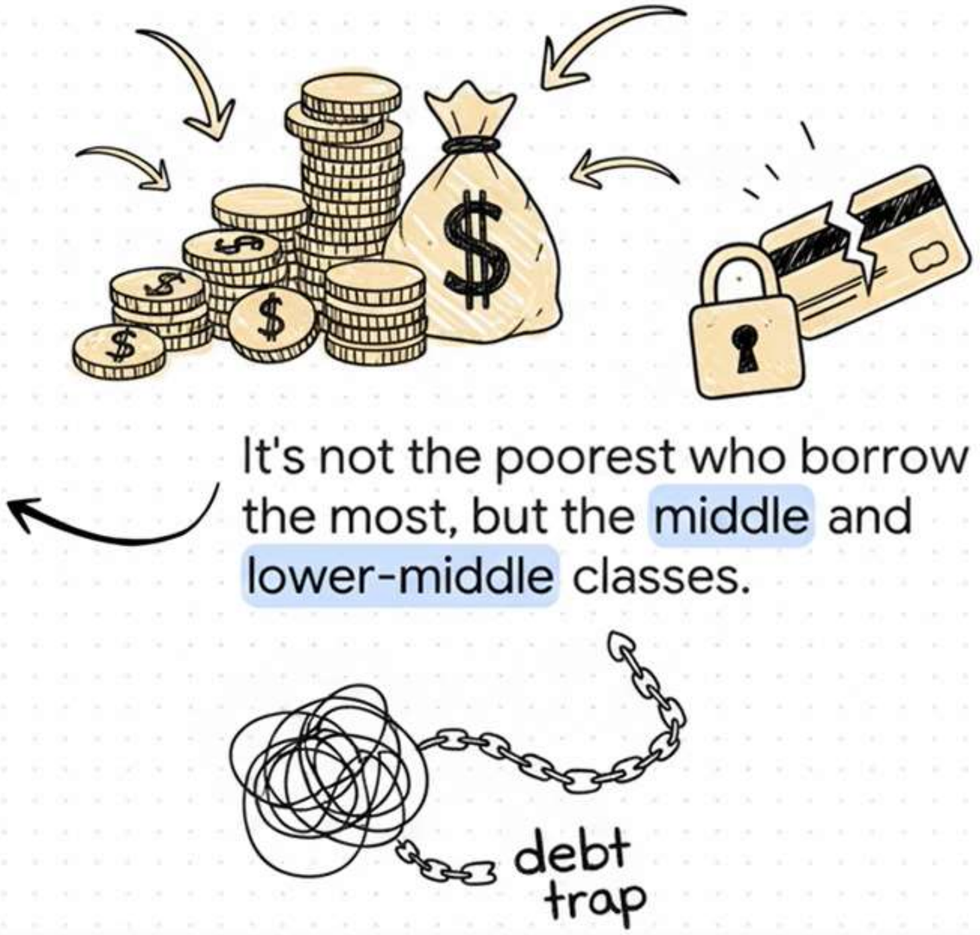
Borrowing peaks in mid-life, with the 35-44 age group showing the highest rate at nearly 75%.



Progress - Key Findings

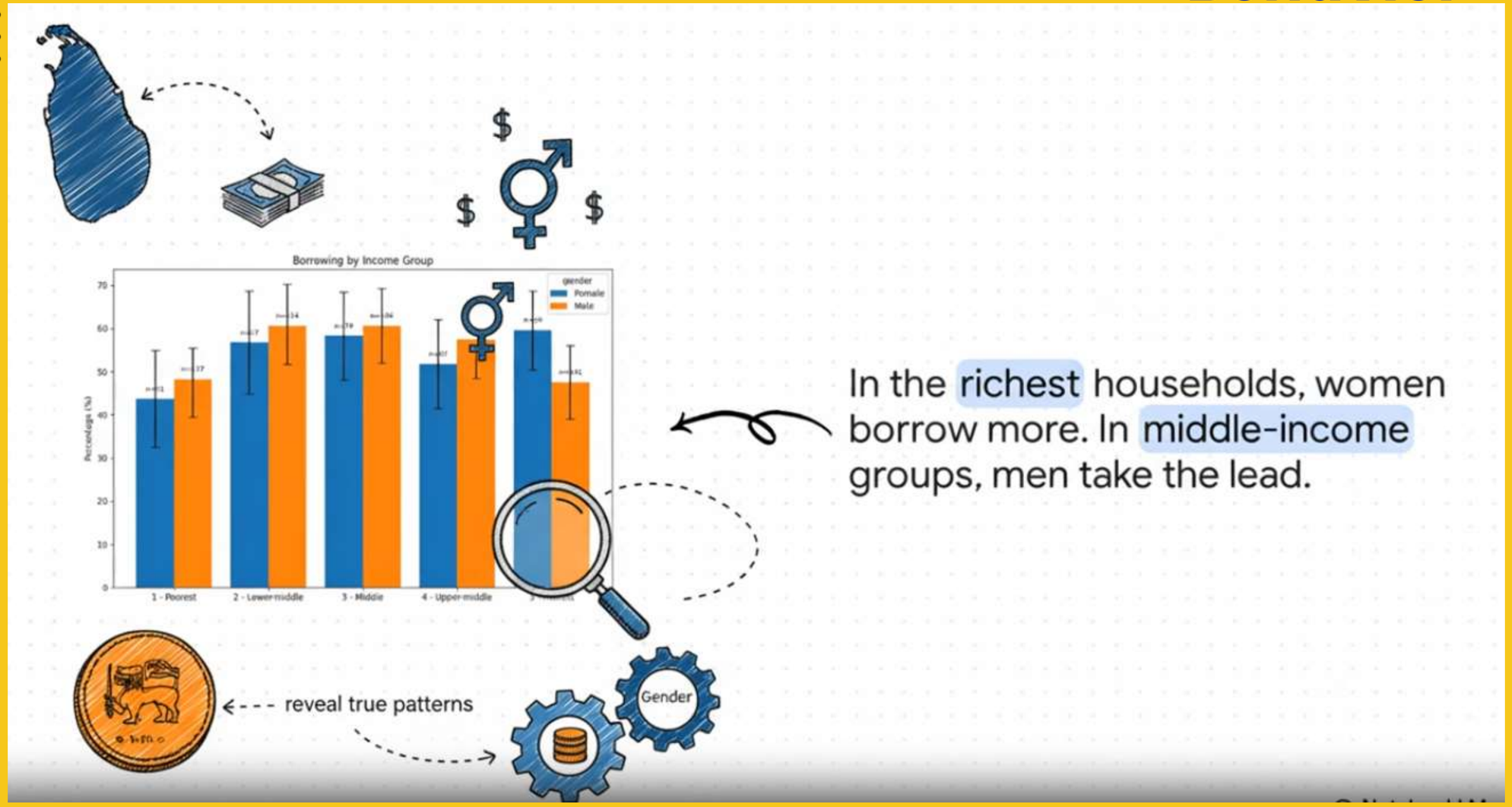
Borrowing Behavior





Progress - Key Findings

Borrowing Behavior



Progress - Key Findings

Borrowing Behavior

Nearly half of all loans are for business/investment, indicating high entrepreneurial activity.

Predict likelihood of borrowing using demographic features in Sri Lanka

STEPS WE FOLLOWED

Predict likelihood of borrowing using demographic features in Sri Lanka

FINAL MODEL SUMMARY

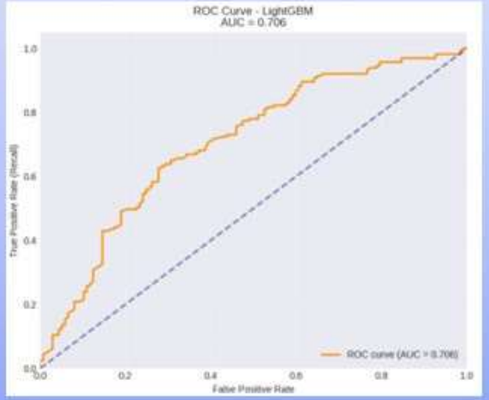
MODEL
LightGBM

Accuracy
0.6500

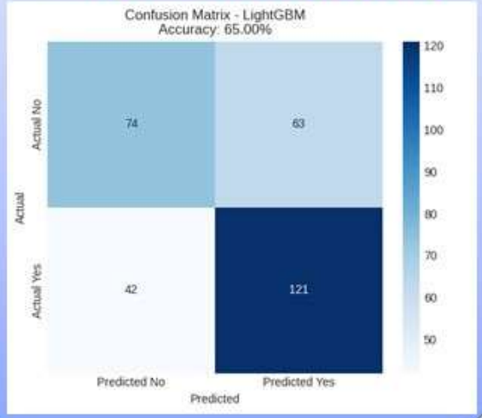
Precision
0.6576

Recall
0.7423

ROC CURVE



CONFUSION MATRIX

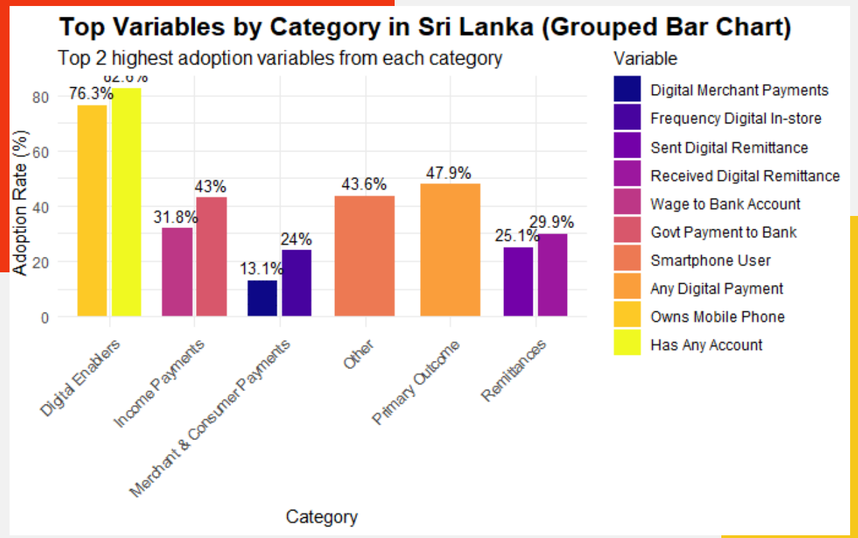


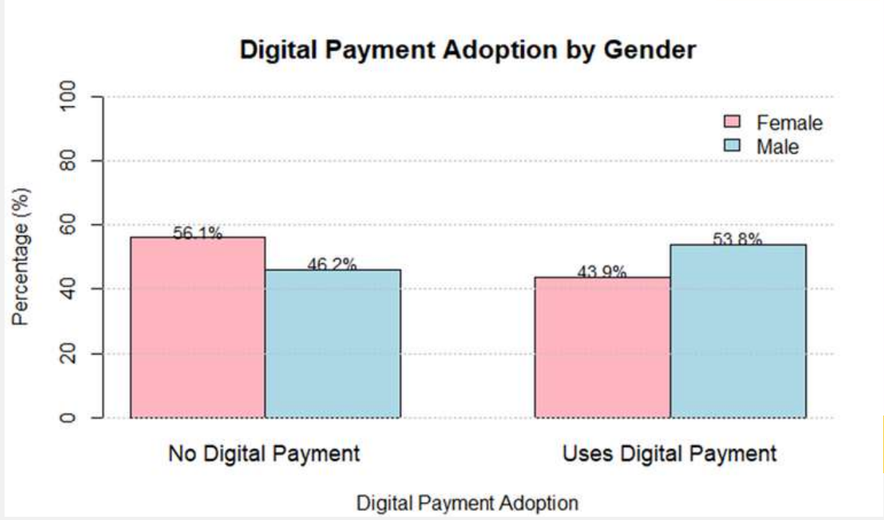
VARIABLES

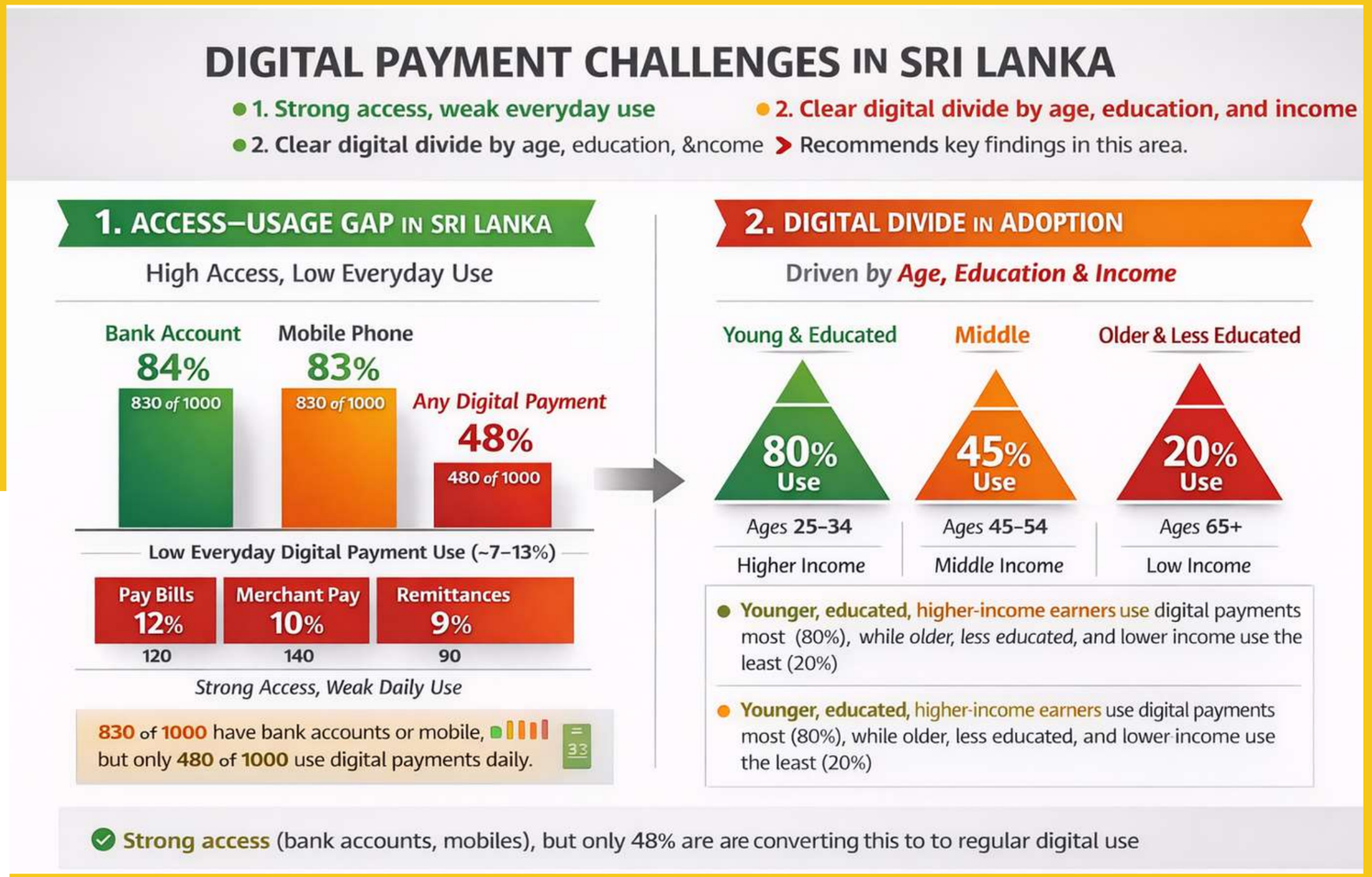
- emp_in
- Educ
- internet_use
- Anydigpayment
- Female
- pay_utilities
- inc_q
- Age

Progress - Key Findings

Digital Payment Adoption







KEY FINDINGS

Interesting findings
(anomalies, opportunities, risks)

- Despite high access, everyday digital payment usage is low (below 15%)
- **Mobile money** usage is almost zero for income payments (<1%), even though ownership is widespread
- Urban-rural adoption gap is minimal, contradicting the common assumption that infrastructure access is the main barrier
- **Younger, educated, high-income** groups show very high adoption, indicating a ready market for advanced digital

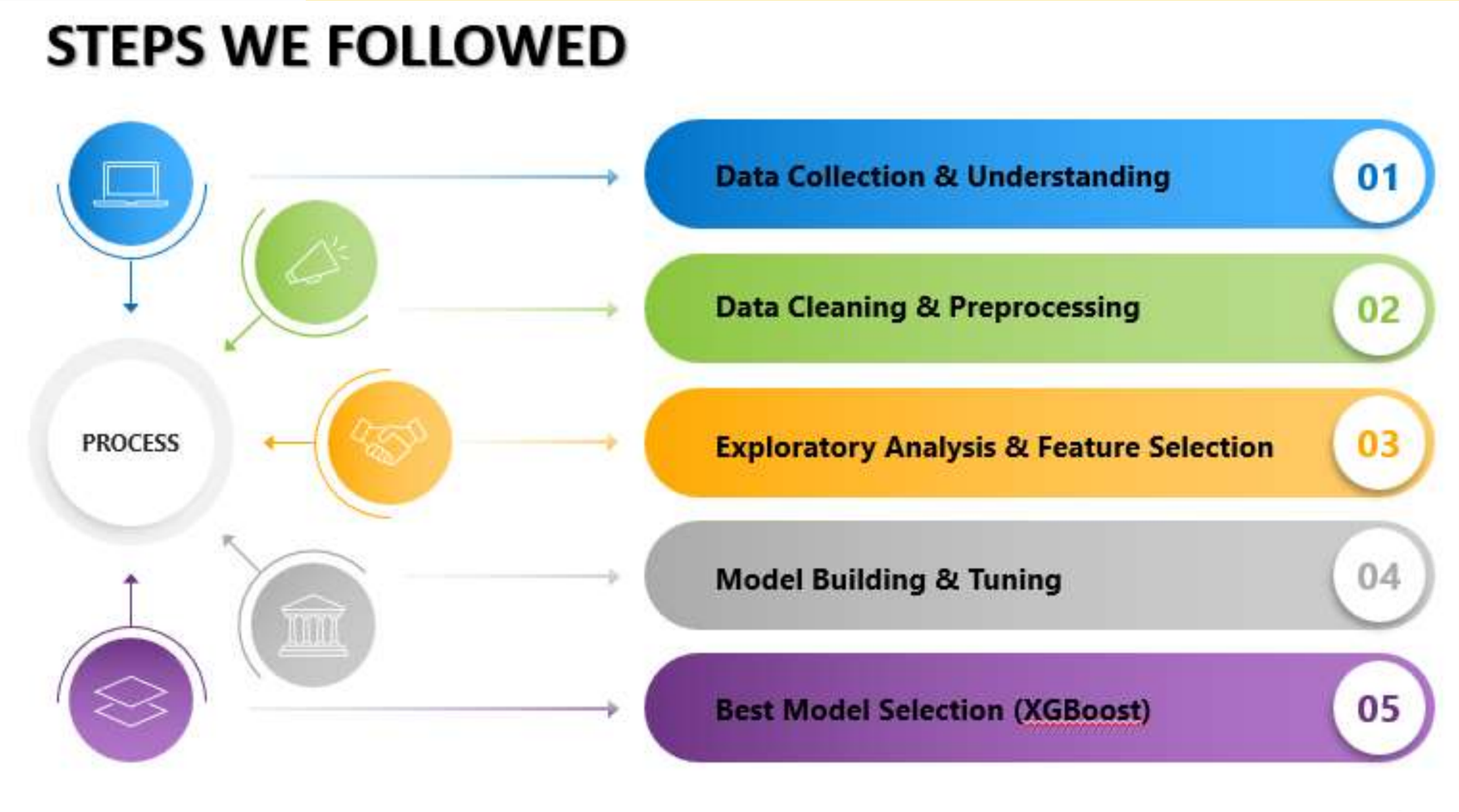
Visual Support Data

- Overall adoption: **48%** use digital payments
  Access vs Usage: **52%** are non-users
- Bank accounts: **84%** Bank accounts
  Mobile phones: **83%** Mobile phones
- Demographic Divide: Ages **25-34**: ~62% adoption
  Ages **60+** ~15% adoption
- Digital Enablers:
  Internet access: **89%**
- Govt. payments: bank: **~38%**
- Merchant payments: bills, remittances: **~20%**

Implications for business

- **Focus on habit formation**
  Encourage frequent, low-value transactions (retail, transport, utities)
- **Incentives** such as discounts, cashback, loyalty programs can shift benvior
- **Target high-income, younger users:** Advanced products, credit, investments
- **Mobile-based payments** for income are almost nonexistent, showing reliance on traditional banking rather than habitual
- **Segment the market strategically:**
  - **High-income**, young users – advanced **fintec**: products, credit, investments

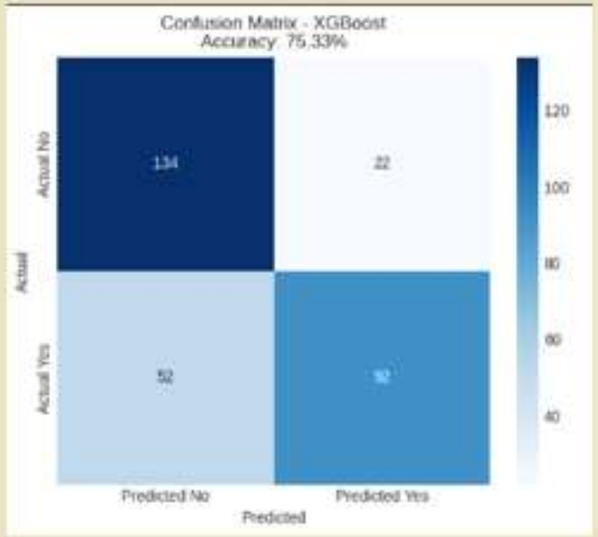
Predicting Digital Payment Adoption in Sri Lanka Using Machine Learning



Predicting Digital Payment Adoption in Sri Lanka Using Machine Learning

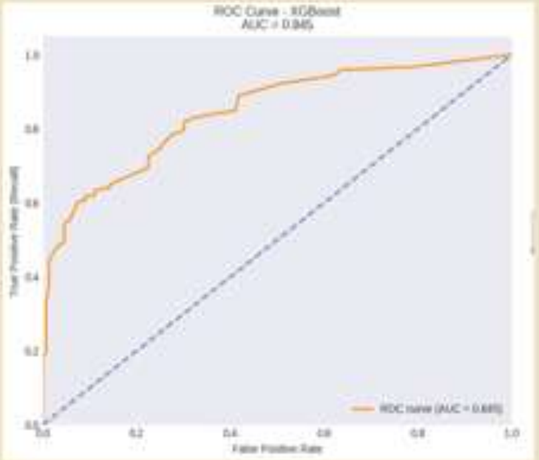
FINAL MODEL SUMMARY

CONFUSION MATRIX



MODEL	Accuracy	Precision	Recall
XGBoost	75.33%	0.8070	0.6389

ROC CURVE

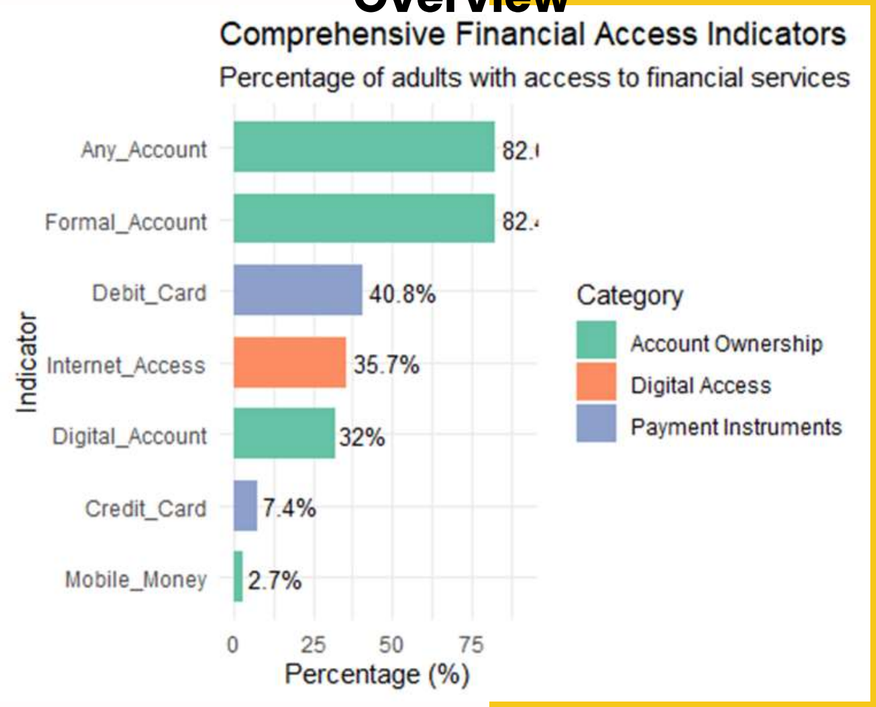


SELECTED VARIABLES

- pay utilities
- Age
- Saved
- Fin32
- fin30
- fin2

Progress - Key Findings

Financial Access Overview



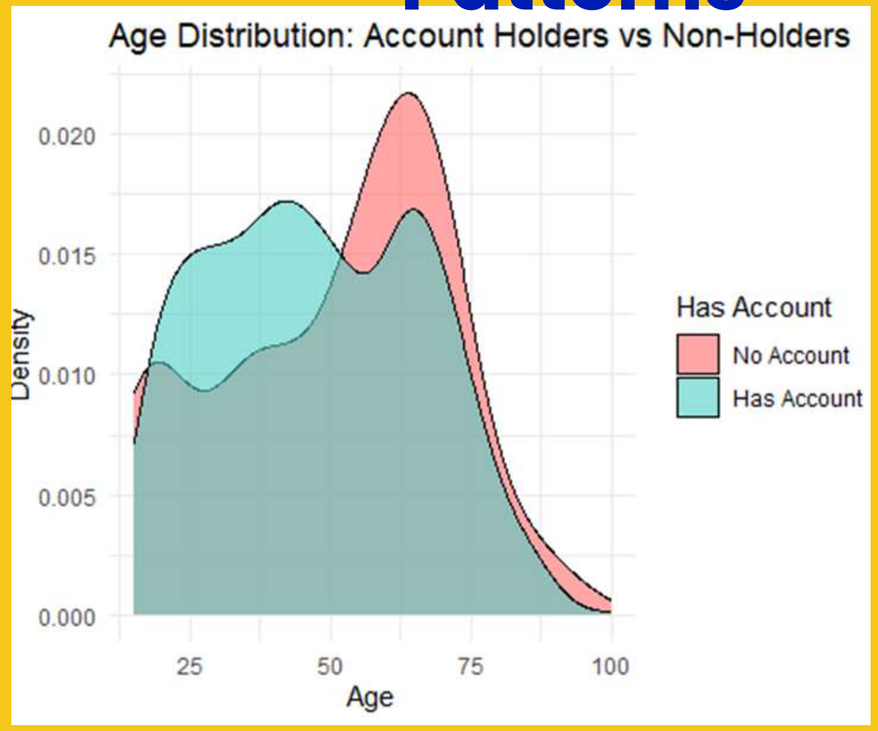
🏦 **Account Ownership 82.6% of adults**

Formal Accounts 82.4%

💳 **Debit Card Usage 40.8%**

📱 **Digital Access 35.7%**

Financial Inclusion Patterns

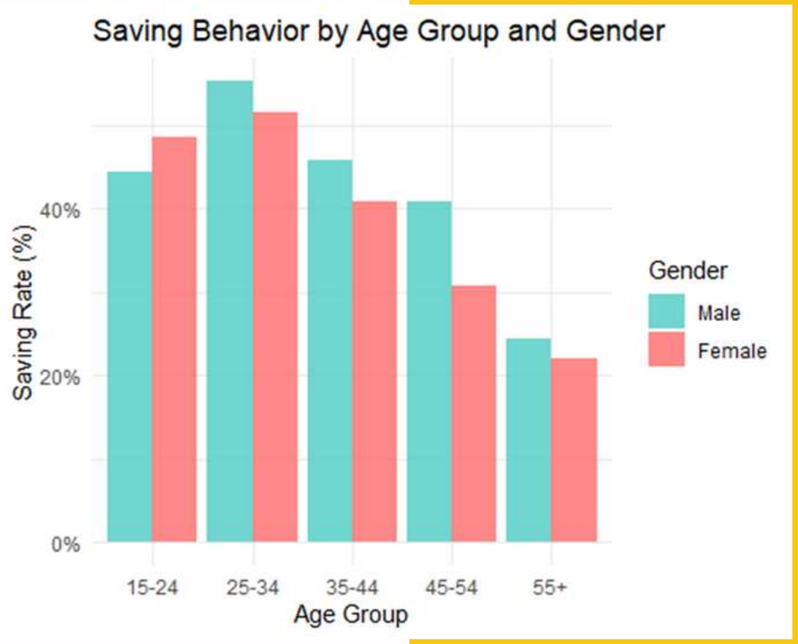


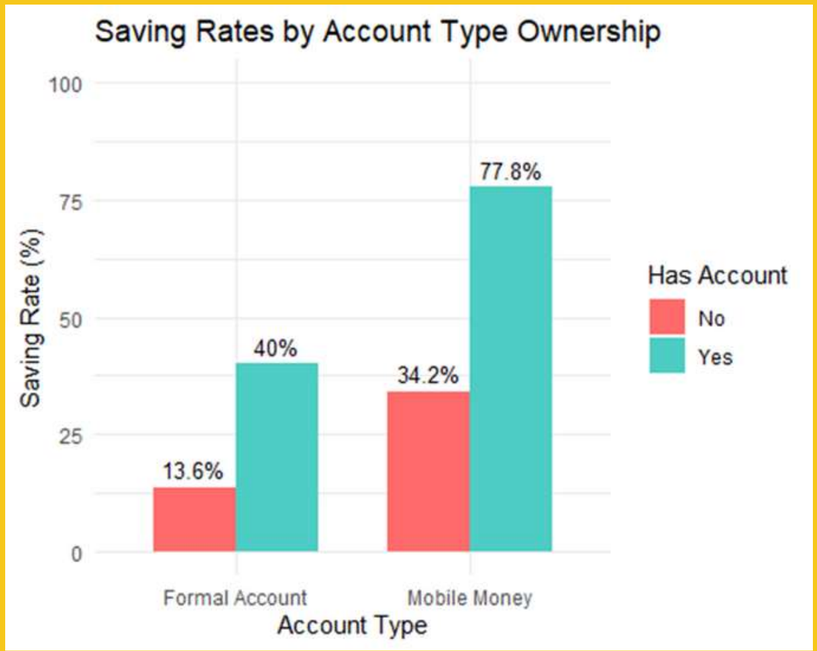
👥 **Account ownership peaks among individuals aged 25–55.**

Progress - Key Findings

Financial Inclusion Patterns

Saving Behavior Overview





💰 **Savings are highest among individuals aged 25–34.**

FUTURE

Finish Clustering (5 days left)

- Group people into 4 or more types
- Name each group
- Give each group different help

Combine All Findings (Next week)

- Put digital payment results + borrowing results together
- Find big patterns
- Make action plan

- Write Final Report
- Design poster
- Practice presentation
- Final submission

Conclusion

- **High Access, Low Use**

Many people have bank accounts, but very few use digital payments daily.

- **Predicting Who Will Adopt**

Our model can predict digital payment adoption with 75% accuracy.

- **Predict Who Will Borrow**

A model can predict borrowing likelihood with 65% accuracy.

- **Mobile Money is Rarely Used for Income**

Almost no one gets paid or makes payments with mobile money.

- **Young, High-Income Users Are Key**

People who are younger, educated, and have higher incomes are the main digital payment users.

Key References

1. Gunasinghe, C., & Perera, L. D. (2022). Financial Inclusion in Sri Lanka: A Demand-Side Analysis. Sri Lanka Journal of Economic Research, 9(1), 33-56.
2. Withana, P., & Rathnayaka, R. M. K. T. (2021). Determinants of digital payment adoption: An empirical study based on UTAUT model among young adults in Sri Lanka. Kelaniya Journal of Management, 10(2), 1-19.
3. Central Bank of Sri Lanka. (2021). *National Financial Inclusion Strategy of Sri Lanka 2021-2025.*
4. International Finance Corporation (IFC). (2020). Digital Financial Services in Sri Lanka: The Way Forward.

Thank You

By Group 04